

Rules of Inference

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5.1 Motivation

An *argument* is a sequence of propositions: *premises* $p_1, \dots, p_k$ followed by a *conclusion* q . The argument is *valid* when the premises entail the conclusion: $(p_1 \wedge \dots \wedge p_k) \rightarrow q$ is a tautology. Validity concerns form, not content — a valid argument with true premises is called *sound* and its conclusion must be true.

5.1.1 The limits of truth tables

Validity can in principle be checked by a truth table, but an argument with n atomic propositions requires 2^n rows: 20 atoms already mean more than a million rows. Rules of inference replace this exponential check by short chains of certified elementary steps — the same economy that proofs provide throughout mathematics.

5.2 The rules of inference

Each rule below is a small valid argument, verified once by a truth table and then reused forever. We write premises above the line, conclusion below.

Rule	Premises	Conclusion	Tautology
Modus ponens	$p, \quad p \rightarrow q$	q	$(p \wedge (p \rightarrow q)) \rightarrow q$
Modus tollens	$\neg q, \quad p \rightarrow q$	$\neg p$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$
Hypothetical syllogism	$p \rightarrow q, \quad q \rightarrow r$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
Disjunctive syllogism	$p \vee q, \quad \neg p$	q	$((p \vee q) \wedge \neg p) \rightarrow q$
Addition	p	$p \vee q$	$p \rightarrow (p \vee q)$
Simplification	$p \wedge q$	p	$(p \wedge q) \rightarrow p$
Conjunction	$p, \quad q$	$p \wedge q$	$((p) \wedge (q)) \rightarrow (p \wedge q)$
Resolution	$p \vee q, \quad \neg p \vee r$	$q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$

Modus ponens is the engine of deduction; resolution alone underlies automated theorem provers and SAT solvers. Beware the two fallacies of Chapter 2: affirming the consequent (q and $p \rightarrow q$, hence p) and denying the antecedent ($\neg p$ and $p \rightarrow q$, hence $\neg q$) are not valid rules.

Example 5.1. Premises: “If it rains, the match is cancelled” ($r \rightarrow c$); “If the match is cancelled, the tickets are refunded” ($c \rightarrow t$); “The tickets are not refunded” ($\neg t$). Conclusion: “It does not rain” ($\neg r$).

Step	Formula	Justification
1	$r \rightarrow c$	premise
2	$c \rightarrow t$	premise
3	$r \rightarrow t$	hypothetical syllogism, 1, 2
4	$\neg t$	premise
5	$\neg r$	modus tollens, 3, 4

Example 5.2. Premises: $p \vee q, \quad p \rightarrow r, \quad q \rightarrow s, \quad \neg r$. Show: s .

Step	Formula	Justification
1	$p \rightarrow r$	premise
2	$\neg r$	premise
3	$\neg p$	modus tollens, 1, 2
4	$p \vee q$	premise
5	q	disjunctive syllogism, 4, 3
6	$q \rightarrow s$	premise
7	s	modus ponens, 6, 5

5.3 Rules for quantified statements

Four rules connect quantifiers with concrete instances. Here c denotes an element of the domain.

Rule	From	Infer
Universal instantiation (UI)	$\forall x P(x)$	$P(c)$ for any c
Universal generalisation (UG)	$P(c)$ for <i>arbitrary</i> c	$\forall x P(x)$
Existential instantiation (EI)	$\exists x P(x)$	$P(c)$ for some <i>fresh</i> name c
Existential generalisation (EG)	$P(c)$ for some c	$\exists x P(x)$

Two safety conditions: in UG, c must be arbitrary — no special assumption may have been made about it; in EI, the witness name c must be new, not previously used in the proof. Violating either produces plausible-looking nonsense.

Example 5.3 (Socrates). Premises: $\forall x (M(x) \rightarrow D(x))$ (“all men are mortal”) and $M(\text{socrates})$. By UI, $M(\text{socrates}) \rightarrow D(\text{socrates})$; by modus ponens, $D(\text{socrates})$: Socrates is mortal. Propositional logic alone could never perform the first step.

Example 5.4 (Existential reasoning). Premises: “Some student in the class has not done the homework” $\exists x (S(x) \wedge \neg H(x))$; “Every student in the class passed the test” $\forall x (S(x) \rightarrow T(x))$. Conclusion: “Someone who passed the test has not done the homework” $\exists x (T(x) \wedge \neg H(x))$. Proof: by EI choose a fresh c with $S(c) \wedge \neg H(c)$; by simplification $S(c)$ and $\neg H(c)$; by UI on the second premise, $S(c) \rightarrow T(c)$; by modus ponens $T(c)$; by conjunction $T(c) \wedge \neg H(c)$; by EG, $\exists x (T(x) \wedge \neg H(x))$.

5.4 Summary

Valid argument forms — modus ponens, modus tollens, syllogisms, addition, simplification, conjunction, resolution — replace exponential truth-table checks by linear chains of certified steps. The four quantifier rules (UI, UG, EI, EG) let deductions pass between general statements and concrete instances, with the side conditions “arbitrary” (UG) and “fresh” (EI). Together these rules constitute the deductive machinery used implicitly by every proof in the remainder of the book.

Exercises

Exercise 5.1 EASY

Name the rule of inference used: (a) “It is snowing, therefore it is snowing or it is cold.” (b) “It is snowing and cold; therefore it is cold.” (c) “If I sleep, I miss the bus. I slept. Therefore I missed the bus.”

Exercise 5.2 EASY

Verify by truth table that disjunctive syllogism $((p \vee q) \wedge \neg p) \rightarrow q$ is a tautology.

Exercise 5.3 EASY

Identify the fallacy: “If the program is correct, all tests pass. All tests pass. Therefore the program is correct.”

Exercise 5.4 MEDIUM

Construct a step-by-step deduction. Premises: $\neg p \rightarrow q$, $q \rightarrow r$, $\neg r$. Conclusion: p .

Exercise 5.5 MEDIUM

Construct a deduction. Premises: $p \rightarrow (q \wedge r)$, $p \vee s$, $\neg q$. Conclusion: s .

Exercise 5.6 MEDIUM

Use resolution repeatedly to derive a contradiction (the empty clause) from the clauses: $p \vee q$, $\neg p \vee q$, $\neg q$.

Exercise 5.7 MEDIUM

Premises: “Every computer science student studies discrete mathematics” and “Hind is a computer science student”. Formalise with predicates and derive “Hind studies discrete mathematics”, naming each rule.

Exercise 5.8 HARD

Find the flaw: from $\exists x P(x)$ and $\exists x Q(x)$, a student instantiates both with the same c to get $P(c) \wedge Q(c)$ and concludes $\exists x (P(x) \wedge Q(x))$. Exhibit concrete P , Q and a domain where the premises are true and the conclusion false.

Exercise 5.9 HARD

Premises: $\forall x (P(x) \rightarrow Q(x))$ and $\forall x (Q(x) \rightarrow R(x))$. Give a full quantified deduction of $\forall x (P(x) \rightarrow R(x))$, naming every rule (UI, UG, hypothetical syllogism).

Solutions to the Exercises

Solution 5.1. (a) Addition. (b) Simplification. (c) Modus ponens. ■

Solution 5.2. The formula fails only if $(p \vee q) \wedge \neg p$ is true and q false. Then p is false (from $\neg p$) and q false, making $p \vee q$ false — contradiction. So no falsifying row exists: a tautology. (The four-row table confirms: rows (T, T), (T, F) make $\neg p$ false; row (F, T) makes q true; row (F, F) makes $p \vee q$ false.) ■

Solution 5.3. Affirming the consequent: from $p \rightarrow q$ and q one cannot infer p — tests can pass for an incorrect program. ■

Solution 5.4. 1. $q \rightarrow r$ (premise); 2. $\neg r$ (premise); 3. $\neg q$ (modus tollens, 1,2); 4. $\neg p \rightarrow q$ (premise); 5. $\neg\neg p$, i.e. p (modus tollens, 4,3, with double negation). ■

Solution 5.5. 1. $\neg q$ (premise); 2. $\neg q \vee \neg r$ (addition, 1); 3. $\neg(q \wedge r)$ (De Morgan, 2); 4. $p \rightarrow (q \wedge r)$ (premise); 5. $\neg p$ (modus tollens, 4,3); 6. $p \vee s$ (premise); 7. s (disjunctive syllogism, 6,5). ■

Solution 5.6. Resolve $p \vee q$ with $\neg p \vee q$ on p : get $q \vee q = q$. Resolve q with $\neg q$: the empty clause. The clause set is unsatisfiable. ■

Solution 5.7. Let $C(x)$: “ x is a CS student”, $D(x)$: “ x studies discrete mathematics”. Premises: $\forall x (C(x) \rightarrow D(x))$, $C(\text{hind})$. 1. UI: $C(\text{hind}) \rightarrow D(\text{hind})$. 2. Modus ponens with the second premise: $D(\text{hind})$. ■

Solution 5.8. The error is reusing the name c in the second EI — the witness must be fresh. Counterexample: domain $\mathbb{Z}$, $P(x)$: “ x is even”, $Q(x)$: “ x is odd”. Both premises are true, yet no integer is both even and odd, so $\exists x (P(x) \wedge Q(x))$ is false. ■

Solution 5.9. Let c be arbitrary. 1. UI on premise 1: $P(c) \rightarrow Q(c)$. 2. UI on premise 2: $Q(c) \rightarrow R(c)$. 3. Hypothetical syllogism: $P(c) \rightarrow R(c)$. 4. Since c was arbitrary, UG yields $\forall x (P(x) \rightarrow R(x))$. ■